

Introduction to Modeling for Formal Verification

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Today's goals



▼

Module 1

▼

Introduction to Model Checking

Module 2

▲

Modeling

Background

✓

Propositional Logic

Video • 13 min

✓

Predicate Logic

Video • 9 min

✓

Set Theory

Video • 11 min

✓

Foundations of Logic and Set Theory - Key Concepts and Applications

Reading • 1h

✓

Background Quiz

Graded Assignment • Grade: 100%

Transition Systems

✓

Transition Systems: Definitions

Video • 13 min

✓

Deterministic vs Non-Deterministic Systems

Video • 8 min

✓

Executions

Video • 5 min

✓

Modeling Sequential Circuits

Video • 11 min

✓

Principles of Model Checking - Section 2.1, Page 19-29

Reading • 45 min

✓

Transition Systems Quiz

Graded Assignment • Grade: 100%

Program Graphs

✓

Program Graphs: Definitions

Video • 9 min

✓

Program Graphs: Examples

Video • 15 min

✓

From Program Graphs to Transition Systems

Video • 11 min

✓

Principles of Model Checking - Section 2.1, (Page 29-34)

Reading • 30 min

●

Program Graphs Quiz

Graded Assignment • 15 min

Module 3

▼

Modeling System Composition

Principles of Model Checking - Section 2.1, (Page 29-34)



0:05 / 0:32

2x





Textbook - Principles of Model Checking - Chapter 2.2 (page 29-34)

Pages 29-33 provide definitions for program graphs and explain how a new version of the beverage vending machine example is modeled using a program graph. The method for finding the underlying transition systems for program graphs is described on pages 33-35. Figure 2.3 shows the corresponding transition system for the program graph of the new beverage vending machine as given in Example 2.12.

Go to next item

✓ Completed

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